BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Tenth, 2022, Lahore Board) (Objective Type) (Group - I)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. PAN stands for

(A) Poly aniline nitrate
(B) Peroxy acetyl nitrate
(C) Poly acetyl nitrate
(D) Polyoxy acetyl nitrate

Q2. Which one of the following ions does not cause hardness in water:

(A) Ca^{2+}
(B) Mg^{2+}
(C) SO_4^{2-}
(D) Na^{+}

Q3. Which one of the following is synthetic fibre

(A) Cotton
(B) Silk
(C) Wool
(D) Nylon

Q4. The envelope of different gases around the earth is called:

(A) Atmosphere
(B) Biosphere
(C) Lithosphere
(D) Hydrosphere

Q5. Red hair contain compound of:

(A) Iron
(B) Copper
(C) Titanium
(D) Molybdenum

Q6. Which gas is used to prepare ammonia:

(A) N_2 (B) O_2

(C) Cl_2 (D) SO_2

Q7. The sources of galactose are:

(A) Fruits (B) Rice

(C) Cotton (D) Cellulose

Q8. Dilute acids react with carbonates to produce the given product except:

(A) Salt (B) Water

(C) Carbon dioxide (D) Hydrogen

Q9. Main cause of global warming is:

(A) O_2 gas (B) N_2 gas

(C) CO_2 gas (D) CO gas

Q10. Which one of these is a saturated hydrocarbon:

(A) C_2H_4 (B) C_3H_6

(C) C_4H_8 (D) C_5H_{12}

Q11. The example of heterocyclic compound is:

(A) Benzene (B) Hexane

(C) Cyclohexane (D) Pyridine

Q12. Water has maximum density at:

(A) 10°C (B) 4°C

(C) 0°C (D) 100°C

Chemistry (Tenth, 2022, Lahore Board) (Essay Type) (Group - I)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

QPART - I.

QQ.2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) If reaction quotient Qc of a reaction is more than Kc, what will be the direction of the reaction?

(ii) What are monosaccharides? Give one example.

(iii) What is ammonical liquor?

(iv) Write down the two characteristics of homologous series.

(v) Define organic compound with an example.

(vi) What is coal tar?

(vii) Describe the sources and uses of vitamin A.

(viii) Why are the ten amino acids essential for us?

QQ.3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Name two acids used in the manufacturing of fertilizers.

(ii) How are soluble salts recovered from water?

(iii) Write down two uses of ethene.

(iv) Why are the alkanes called paraffins?

(v) What is the principle of Solvay's process?

(vi) Write down names and formulae of two copper ores.

(vii) Write two uses of fuel oil.

(viii) What is roasting?

QQ.4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write composition of air in dry state.

(ii) What is ozone hole? Where it was noticed first?

(iii) Why does temperature decrease in troposphere?

(iv) How acid rain increases acidity of soil?

(v) How does limestone dissolve in water?

(vi) Compare soft and hard water.

(vii) Why are pesticides used?

(viii) How boiling process removes hardness from water? Give reaction.

Chemistry (Tenth, 2022, Lahore Board) (PART-II) (Group - I)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt TWO questions.

QQ.5(a). (Marks 5) Explain the sources and uses of proteins.

QQ.5(b). (Marks 4) Determine the concentration units for the following reactions: (i) $H_2 + I_2 \rightleftharpoons 2HI$ (ii) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

QQ.6(a). (Marks 5) Give uses of five different acids in every day life.

QQ.6(b). (Marks 4) Define hard water. Give three disadvantages of hard water.

QQ.7(a). (Marks 5) Define concentration of ores. Explain gravity separation and froth floatation process of ore concentration.

QQ.7(b). (Marks 4) Describe the physical properties of alkynes.

GROUP – II

Chemistry (Tenth, 2022, Lahore Board) (Objective Type) (Group - II)

Time Allowed: 15 Min. | **Maximum Marks:** 12

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Formula of ozone is:

(A) O_2 (B) O_3

(C) O (D) CO

Q2. Photosynthesis produces:

(A) Starch (B) Glucose

(C) Sucrose (D) Cellulose

Q3. Urea contains nitrogen:

(A) 45.6% (B) 44.6%

(C) 46.6% (D) 47.6%

Q4. Kc depends upon:

(A) Temperature (B) Pressure

(C) Volume (D) Atmosphere

Q5. Permanent hardness is removed by adding:

(A) Na_2 Zeolite (B) Soda lime

(C) Limewater (D) Quicklime

Q6. Which one of the following is amphoteric:

(A) BF_3 (B) NaCl

(C) H_2O (D) SO_4^{2-}

Q7. Which of the following gas is used to destroy harmful bacteria in water:

(A) Iodine
(B) Fluorine
(C) Chlorine
(D) Bromine

Q8. The functional group -COOH is found in:

(A) Carboxylic acids
(B) Aldehydes
(C) Alcohols
(D) Esters

Q9. Main compound of natural gas is:

(A) Propane
(B) Methane
(C) Butane
(D) Propene

Q10. Which gas protects the earth's surface from ultraviolet radiations:

(A) CO_2
(B) CO
(C) N_2
(D) O_3

Q11. Dehydration of alcohols can be carried out with:

(A) NaOH
(B) KOH
(C) HCl
(D) H_2SO_4

Q12. Which is a secondary pollutant:

(A) SO_2
(B) CO_2
(C) CH_4
(D) H_2SO_4

Chemistry (Tenth, 2022, Lahore Board) (Essay Type) (Group - II)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

QPART - I.

QQ.2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write the equilibrium constant expression for the reaction: $CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$.

(ii) How dynamic equilibrium is established?

(iii) What is catenation? Give an example.

(iv) Write two different radicals of butane.

(v) Define functional group with an example.

(vi) Give any two general characteristics of organic compounds.

(vii) What are monosaccharides? Give one example.

(viii) Where are proteins found?

QQ.3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Find out the POH of 0.001 M solution of KOH.

(ii) Give the two uses of calcium chloride?

(iii) Define unsaturated hydrocarbons with example.

(iv) Write the uses of chloroform and carbon tetrachloride.

(v) What are ores? Give an example.

(vi) How carbonation of ammonical brine is carried out in Solvay's process? Write equation.

(vii) Write down the composition and boiling range of kerosene oil.

(viii) What is difference between slag and matte?

QQ.4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Why CO_2 is called a green house gas?

(ii) Write down two sources of oxides of carbon.

(iii) Why should catalytic converters be used in automobile exhaust?

(iv) How does aluminium harm the fish?

(v) Describe occurrence of water.

(vi) Describe chemistry of swimming pool cleanliness.

(vii) How heavy metals are toxic and health hazards for human being?

(viii) Describe chemistry of removing hardness of water by Clark's method.

Chemistry (Tenth, 2022, Lahore Board) (PART-II) (Group - II)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt TWO questions.

QQ.5(a). (Marks 5) What are amino acids? Describe how they are building blocks of proteins?

QQ.5(b). (Marks 4) Define equilibrium constant and what is its importance in predicting extent of a reaction.

QQ.6(a). (Marks 5) Describe any five chemical properties of acid.

QQ.6(b). (Marks 4) Discuss the dual effects of agricultural effluents.

QQ.7(a). (Marks 5) How is crude oil refined? Explain two important fractions of petroleum along with their usage.

QQ.7(b). (Marks 4) Write four uses of acetylene.